

Hongyu Liu

📞 13538417815

✉ david.liu1888888@gmail.com

🏠 [Personal Website](#)

🌐 [Research Blog](#)

Education

The Chinese University of Hong Kong, Shenzhen

09/2026 – 06/2028 (Expected)

Master's in Data Science

South China Normal University

09/2022 – 06/2026

Bachelor's in Software Engineering

GPA: 4.06/5 (Top 10.5%)

Technology Trends and Industry Analysis

Reasoning Compute in the Agent Era

[Blog](#)

Maps GPU/ASIC, HBM, packaging, data-center, and cloud profit pools by effective task cost; tracks CapEx, contracts, financing, rents, utilization, renewals, and returns to spot overheating and profit shifts.

AI Agents and Enterprise Coordination

[Blog](#)

Compares enterprise agent approaches from Grok Bot, Claude Tag, and Doubao Work Partner across context, permissions, handoffs, and acceptance; examines value in collaboration entry points and cross-system governance, with indicators for enterprise willingness to pay.

Safety Gaps in Speech LLMs

[Blog](#)

Extends VoxPrivacy and VoxSafeBench research into Speech AI industry analysis, covering model capability, safety evaluation, and vertical applications; identifies potential opportunities in continuous evaluation infrastructure and high-value use cases.

Research and Projects

Trustworthy Speech LLM Evaluation

07/2025 – 01/2026

CUHK-Shenzhen Speech Team Collaboration · Second Author · Benchmark Design & Build · Data Processing

[VoxPrivacy](#) · [ICLR](#)
[VoxSafeBench](#)

- Built two benchmarks spanning multi-user interaction privacy, safety, and fairness. Cross-model evaluation found that frontier models detect speaker, tone, and environmental cues but struggle to use them reliably in privacy and safety decisions, exposing a gap between hearing cues and making sound decisions.

Audio Dialogue and Intent Recognition

02/2025 – 06/2025

First Author · Data Processing · Model Design · Experimental Validation

[DialogGraph-LLM](#) · [ECAI Oral](#)
[MSMT-FN](#) · [ADMA](#)

- Studied intent recognition in long, multi-speaker dialogue through multi-segment, multi-task fusion, relational dialogue modeling, and semi-supervised learning; evaluated how different approaches map raw audio to intent predictions.

TradingAgent | Multi-Agent Investment Research and Thesis Validation Framework

01/2026 – Present · [GitHub](#)

Restructured multi-agent analysis for A-share company research into a unified evidence chain across filings, financials, news, markets, and valuation; applied evidence checks, bull/bear cross-validation, and historical and peer valuation constraints to produce traceable theses, risks, and validation criteria

Internship Experience

Insta360

02/2026 – 06/2026

Speech Algorithm Intern, Research Institute · Shenzhen

- **On-Device Speech Products and Model Evaluation:** Compared cascaded and end-to-end speech models for drone and assistant intent tasks, keeping models under 50M parameters at baseline; evaluated multimodal LLMs on image understanding, answer quality, simultaneous interpretation, and tool use.

Amphion Technology Co., Ltd.

06/2025 – 09/2025

R&D Department · Shenzhen

- **Speech Product Development:** Developed core algorithms for Makaw, a voice-cloning app, including model training and business iterations; contributed ASR/TTS to a video-translation backend.

Ruiju Intelligent Technology Co., Ltd.

07/2024 – 08/2024

AI R&D Group · Guangzhou

- **Purchase Intent Recognition:** Built a model using telemarketing call audio to identify customer purchase intent and assess conversion propensity.

Skills and Honors

Honors: Provincial Second Prize, China Undergraduate Mathematical Contest in Modeling (2024) | Honorable Mention, MCM (2025) | National Second Prize, MathorCup (2024)

Skills: Python for data analysis | Agent tools including Codex and Claude Code | English (IELTS 6.5)

Activities: Staff Member, Youth League Committee Research and Survey Department | College Basketball Team Captain